

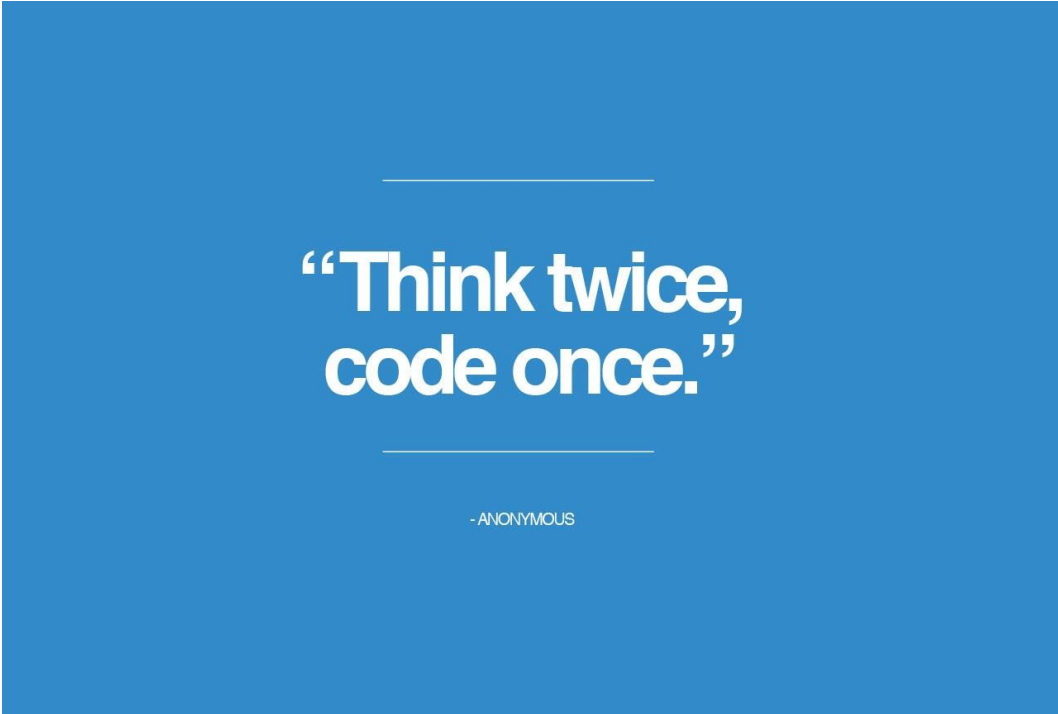


Variables and Constants

CHAPTER 3

Coding the Algorithm

Refers to the process of translating the algorithm into a language that the computer can understand



**“Think twice,
code once.”**

- ANONYMOUS

Begin by---

Assigning names to every

- ❖ input item,
- ❖ processing item and
- ❖ output item.

Then, assign a **data type** and **initial value**.

Internal Memory Definitions

Each memory location has a **unique** internal address.

Each memory location can only hold **one** item.

Declaring a memory location definition

In C++ this is done with **one statement** that declares the name, data type, and initial value.

Names are descriptive because they are easier to remember and help self-document code.

Types of memory locations

Variables and Named Constants



Variables

Memory location whose **value can change during runtime** (when the program is running)



Named constant

Cannot change during runtime

Example: $\text{PI} = 3.14$

Memory locations

Every memory location declared must have a name or identifier that is descriptive

A good name is one that is **meaningful** when you name it and years later.

C++ Rules

- ❖ Begin with a letter
- ❖ Contain only letters, numbers and underscore
- ❖ No punctuation marks, spaces or special characters
- ❖ Cannot be a keyword (reserved word)
- ❖ Names are case sensitive
 - Discount and discount might look alike, but are different in C++

Common guidelines

- ❖ Use **uppercase** letters for named constants
- ❖ Use **lowercase** for variables
- ❖ If the name is more than one word for a **named constant**, use the **underscore** to separate the names.
 - TAX_RATE
- ❖ If the **variable** name has two words, use the **camel case**:
 - grossPay

Why are these invalid names?

2011sales

end Balance

first.name

int

RATE%



Why are these invalid names? Answer

2011sales – starts with a number

end Balance – has a space

first.name – has a decimal

Int – key word

RATE% - cannot use percent



How many items can a memory location store at one time?



Answer: ONE



Which of the following is correct for the quantity variable?

1. quantity
2. QUANTITY
3. Quantity
4. Any of the above

Answer: Which of the following is correct for the quantity variable?

1. **quantity**
2. QUANTITY
3. Quantity
4. Any of the above

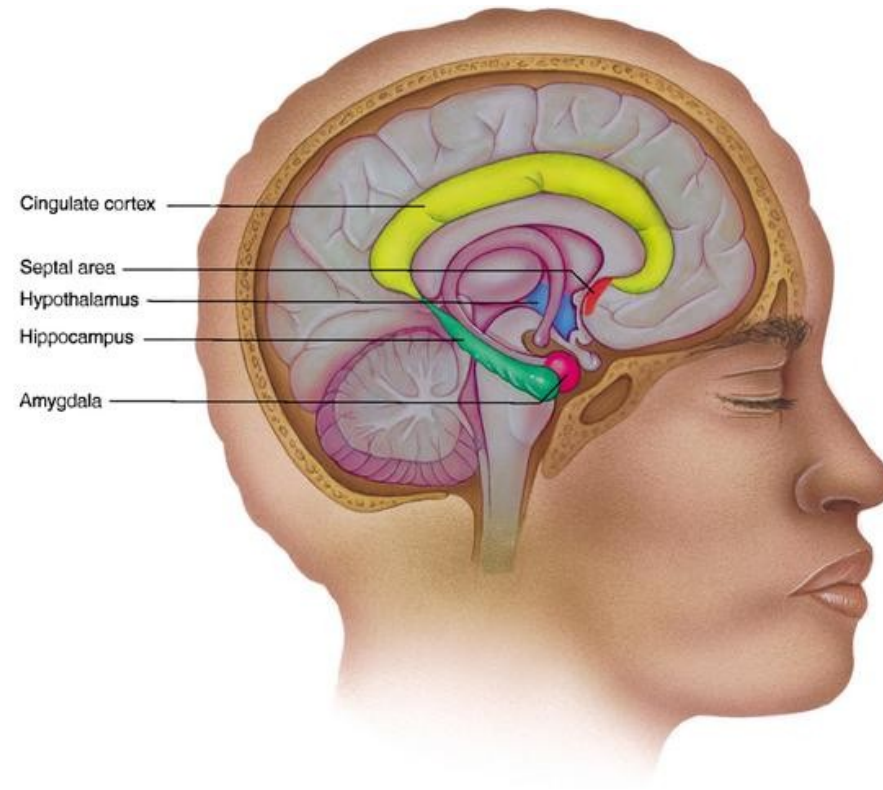
Which of the following is a valid name for a memory location?

1. jan.Sales
2. 2ndQuarterIncome
3. COMMISSION_RATE
4. march\$

Answer: Which of the following is a valid name for a memory location?

1. jan.Sales
2. 2ndQuarterIncome
3. **COMMISSION_RATE**
4. march\$

What are the 2 types of memory locations a programmer can declare?



Answer:

Named Constants and Variables

Fundamental data types in C++

short

int

float

double

bool

char

These are basic types built into C++

Also called primitive or built-in

string data type

User-defined data type

A class: a group of instructions that the computer uses to create an object.

In this case, the class creates a **string** variable which is an object with **zero or more characters**.

char data type

Can only store **one** character like a letter or number or symbol.

It does not store a person's name.



short and int

Store integers only.

A whole number does not contain decimals.

The difference between short and int is the storage size of the variable.

Real numbers

Float

Double

Difference is range.

Double is more precise.

Not all numbers can be represented with accuracy to the penny.

Internal memory

Binary number system

0, 1

ASCII

Contains a **specific numeric code** for each character on the keyboard.

ASCII is then converted to binary.

Data type

Determines how the computer interprets a memory location's existing data.

True or false

The **string** data type is one of the fundamental data types?

Answer: True or **false**

The **string** data type is one of the fundamental data types?

In the binary number system the decimal number 32 is represented as

1. 10000
2. 10001
3. 100000
4. None of the above

Answer: In the binary number system the decimal number 32 is represented as

- 1. 10000
- 2. 10001
- 3. **100000**
- 4. None of the above

What is the **ASCII code** for the lowercase letter b and how is it represented in the computers internal memory?

Answers:

98

01100010

Which data type can store a real number?

- A. double
- B. int
- C. float
- D. Both a and c

Answer: Which data type can store a real number?

- A. double
- B. int
- C. float
- D. Both a and c**

Initial values

Called initializing

bool is done with “true” or “false”

All others use a literal constant

Literal constant

Item of data that can appear in a program instruction and be stored in a memory location.

Must match the data type of the memory location to which it is assigned.

examples

- ❖ Memory locations which are short or int should have integers assigned to them. Example: `int number = 0;`
- ❖ Memory locations which are float or double should have real numbers assigned to them. Example: `double rate = 0.0;`
- ❖ Integers and real numbers are called numeric literal constants.
- ❖ They can contain a minus sign, plus sign, numbers, and the decimal point.
- ❖ Also the letter E for exponential notation

Numeric literal constants

Cannot contain a space, comma, or special character

No \$ or %

A numeric literal constant with no decimal is an int

A numeric literal constant with a decimal is a double data type

Character literal constant

Used to initial char data types

Enclosed in a single quote

- 'x'
- ''

String literal constant

Used for a string memory location

Uses double quotation marks

- "hello"
- "" (no space between) also called empty string

Assigning a value to a memory location

1. Computer compares the values' data type with the memory location's data type
2. If they don't match, then the computer does a **implicit** type conversion
3. Example;
 - IF YOU ASSIGN AN INTEGER 9 TO A REAL NUMBER, THE 9 BECOMES A 9.0
4. When a value is converted from one data type to another that can store LARGER numbers, it is promoted.

demoted

If a int variable is declared and you use a real number to initialize the variable, then the value is demoted:

Example: `int years = 1.5`

Then 1.5 becomes 1 and the value is demoted!

How to initialize:

short and int = 0

double and float = 0.0

char = ' '

string = ""

bool = true or false

Declaring a memory location

Use a statement

```
datatype variablename = initial value;
```

Use the correct syntax! (rules of the programming language)

Examples of initialization

```
int age = 0;
```

```
double price = 0.0;
```

```
bool paid = false;
```

```
char grade = ' ';
```

```
string company = "";
```

Naming a named constant

```
const datatype constantname = value;
```

```
const double PI = 3.14;
```

Which of the following is a character literal constant?

- A. '56'
- B. '%'
- C. "a"
- D. Both a and c

Answer: Which of the following is a character literal constant?

A. '56'

B. '%'

C. "a"

D. Both a and c

Which of the following is a string literal constant?

- A. "\$"
- B. "Good Morning"
- C. ""
- D. All of the above

Answer: Which of the following is a string literal constant?

- A. "\$"
- B. "Good Morning"
- C. ""
- D. **All of the above**

If you assign the number 10.25 to an int variable, what will the computer store in its memory?

Answer:

If you assign the number 10.25 to an int variable, what will the computer store in its memory?

10

Write a C++ statement that declares and initializes an int variable named population.



Answer: Write a C++ statement that declares and initializes an int variable named population.

```
int population = 0;
```

Write a C++ statement that declares the MAX_PAY name constant. It should have double data type and the number 25.55.

Answer: Write a c++ statement that declares the MAX_PAY name constant. It should have double data type and the number 25.55.

```
const double MAX_PAY = 25.55;
```

IF you earn \$10.25 per hour and worked 20 hours, how much less would you be paid if your hourly rate was inadvertently assigned to an int variable?

You make \$5 less!

`double hourlyRate = 10.25`

`Pay = 20 * hourlyRate`

`Pay = 20 * 10.25`

`Pay = 205.00`

`int hourlyRate = 10.25`

`Pay = 20 * hourlyRate`

`Pay = 20 * 10`

`Pay = 200`